

MongoDB Compass - localhost:27017/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1) ✕ + ...

Search connecti

localh... + > ...

- admin
- config
- day2
- local

Welcome mongosh: localhost:27017

```
>_MONGOSH
> use shoppingdb
< switched to db shoppingdb
> db.orders.insertMany([
  { customer_id: "C101",
    name: "Asha",
    items: [
      {product: "Mobile",
        price: 15000
      },
      {product: "Earphones",
        price: 2000
      }
    ]
  },
  {customer_id: "C102",
    name: "Kiran",
    items: [
      {product: "Mobile",
        price: 15000
      },
      {product: "Laptop",
        price: 50000
      }
    ]
  }
])
```

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Welcome mongosh: localhost:27017

```
>_MONGOSH
}
]
},
{customer_id: "C103",
  name: "Neha",
  items: [
    {product: "Earphones",
      price: 2900
    },
    {product: "Laptop",
      price: 59000
    }
  ]
}
]);
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a146e31c795a964f12b4204'),
    '1': ObjectId('6a146e31c795a964f12b4205'),
    '2': ObjectId('6a146e31c795a964f12b4206')
  }
}
```

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Welcome mongosh: localhost:27017 +

> _MONGOSH

```
> //Scenario1 : Mobile Product Revenue Report
> db.orders.aggregate([
  { $unwind: "$items" },
  { $match: { "items.product": "Mobile", "items.price": { $gte: 15000 } } },
  { $group: {
    _id: "Mobile Report",
    totalCustomers: { $sum: 1 },
    totalRevenue: { $sum: "$items.price" },
    avgPrice: { $avg: "$items.price" },
    minPrice: { $min: "$items.price" },
    maxPrice: { $max: "$items.price" },
    customerNames: { $push: "$name" },
    firstCustomer: { $first: "$name" }
  } },
  { $out: "mobile_report" }
]);
<
> //Check result
> db.mobile_report.find().forEach(printjson);
< {
  _id: 'Mobile Report',
  totalCustomers: 2,
  totalRevenue: 30000,
```

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```
> //Scenario 2 : High Value Products Report
> db.orders.aggregate([
  { $unwind: "$items" },
  { $match: { "items.price": { $gte: 15000 }, "items.product": { $in: ["Mobile", "Laptop"] } } },
  { $group: {
    _id: "$items.product",
    totalOrders: { $sum: 1 },
    totalRevenue: { $sum: "$items.price" },
    avgPrice: { $avg: "$items.price" },
    minPrice: { $min: "$items.price" },
    maxPrice: { $max: "$items.price" },
    customerList: { $push: "$name" }
  } },
  { $out: "high_value_products_report" }
]);
<
> //Scenario 3 : Customer Spending Report
> db.orders.aggregate([
  { $unwind: "$items" },
  { $group: {
    _id: "$customer_id",
    customerName: { $first: "$name" },
    totalSpent: { $sum: "$items.price" },
```

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>_MONGOSH

```
maxPrice: { $max: "$items.price" },
customerList: { $push: "$name" }
}},
{ $out: "high_value_products_report" }
});
<
> //Scenario 3 : Customer Spending Report
> db.orders.aggregate([
  { $unwind: "$items" },
  { $group: {
    _id: "$customer_id",
    customerName: { $first: "$name" },
    totalSpent: { $sum: "$items.price" },
    avgPrice: { $avg: "$items.price" },
    minPrice: { $min: "$items.price" },
    maxPrice: { $max: "$items.price" },
    productsBought: { $push: "$items.product" },
    productCount: { $sum: 1 }
  }},
  { $match: { productCount: { $gt: 1 }, totalSpent: { $gte: 30000 } } },
  { $out: "customer_spending_report" }
});
<
shoppingdb> |
```

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CONNECTIONS (1) × + ...

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- local

Welcome mongosh: localhost:27017

```
>_MONGOSH
> use moviedb
< switched to db moviedb
> //CASE STUDY 2 : Movie Ticket Booking System
> //Insert customers
> db.customers.insertMany([
  { _id: 1, name: "Arun", profile: { age: 25, city: "Chennai" } },
  { _id: 2, name: "Priya", profile: { age: 22, city: "Bangalore" } },
  { _id: 3, name: "Rahul", profile: { age: 30, city: "Hyderabad" } },
  { _id: 4, name: "Neha", profile: { age: 28, city: "Mumbai" } },
  { _id: 5, name: "Kiran", profile: { age: 35, city: "Pune" } }
]);
< {
  acknowledged: true,
  insertedIds: {
    '0': 1,
    '1': 2,
    '2': 3,
    '3': 4,
    '4': 5
  }
}
> //Insert movies
> db.movies.insertMany([
```

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- admin
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- day2
- local

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```
>_MONGOSH

> //Insert movies
> db.movies.insertMany([
  { _id: "M1", title: "Inception", genre: "Sci-Fi", price: 300 },
  { _id: "M2", title: "Avatar", genre: "Fantasy", price: 350 },
  { _id: "M3", title: "Interstellar", genre: "Sci-Fi", price: 400 }
]);

< {
  acknowledged: true,
  insertedIds: {
    '0': 'M1',
    '1': 'M2',
    '2': 'M3'
  }
}

> //Insert Bookings
> db.bookings.insertMany([
  { _id: 301, customer_id: 1, movie_id: "M1", show_date: "2026-02-01" },
  { _id: 302, customer_id: 1, movie_id: "M2", show_date: "2026-02-03" },
  { _id: 303, customer_id: 2, movie_id: "M1", show_date: "2026-02-05" },
  { _id: 304, customer_id: 3, movie_id: "M3", show_date: "2026-02-06" },
  { _id: 305, customer_id: 3, movie_id: "M1", show_date: "2026-02-08" },
  { _id: 306, customer_id: 4, movie_id: "M2", show_date: "2026-02-10" },
  { _id: 307, customer_id: 5, movie_id: "M3", show_date: "2026-02-12" },
]);
```

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Data Modeling

CONNECTIONS (1) ✕ + ...

Search connecti

localh... + > ...

- admin
- config
- day2
- local

Welcome mongosh: localhost:27017

```
>_MONGOSH
>
> //Total Bookings per movie
> db.bookings.aggregate([
  { $group: { _id: "$movie_id", totalBookings: { $sum: 1 } } },
  { $sort: { totalBookings: -1 } }
]);
< {
  _id: 'M1',
  totalBookings: 3
}
{
  _id: 'M2',
  totalBookings: 3
}
{
  _id: 'M3',
  totalBookings: 2
}
}
> //Movie-wise Revenue report
> db.movies.aggregate([
  {
    $group: {
      _id: "$title",
```

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Data Modeling

CONNECTIONS (1) ✕ + ...

Search connecti

localh... + > ...

- admin
- config
- day2
- local

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```
>_MONGOSH
>
> //Movie-wise Revenue report
> db.movies.aggregate([
  {
    $group: {
      _id: "$title",
      totalMovies: { $sum: 1 },
      totalRevenue: { $sum: "$price" },
      avgPrice: { $avg: "$price" },
      minPrice: { $min: "$price" },
      maxPrice: { $max: "$price" }
    }
  }
])
< {
  _id: 'Avatar',
  totalMovies: 1,
  totalRevenue: 350,
  avgPrice: 350,
  minPrice: 350,
  maxPrice: 350
}
```

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Data Modeling

CONNECTIONS (1)

Search connecti

localh...

- admin
- config
- day2
- local

```
>_MONGOSH
{
  _id: 'Avatar',
  totalMovies: 1,
  totalRevenue: 350,
  avgPrice: 350,
  minPrice: 350,
  maxPrice: 350
}
{
  _id: 'Inception',
  totalMovies: 1,
  totalRevenue: 300,
  avgPrice: 300,
  minPrice: 300,
  maxPrice: 300
}
{
  _id: 'Interstellar',
  totalMovies: 1,
  totalRevenue: 400,
  avgPrice: 400,
  minPrice: 400,
  maxPrice: 400
}
```

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Data Modeling

CONNECTIONS (1) ✕ + ...

Search connecti

localh... + > ...

- admin
- config
- day2
- local

Welcome mongosh: localhost:27017 +

```
>_MONGOSH
>
> //Customer_wise Spending report
> db.bookings.aggregate([
  {
    $group: {
      _id: "$customer_id",
      totalBookings: { $sum: 1 },
      totalSpent: { $sum: 300 },
      avgSpending: { $avg: 300 },
      bookingsList: { $push: "$movie_id" }
    }
  }
])
< {
  _id: 4,
  totalBookings: 1,
  totalSpent: 300,
  avgSpending: 300,
  bookingsList: [
    'M2'
  ]
}
```

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Data Modeling

CONNECTIONS (1) X + ...

Search connecti

localh... + > ...

- admin
- config
- day2
- local

Welcome mongosh: localhost:27017

>_MONGOSH

```
{
  "_id": 2,
  "totalBookings": 1,
  "totalSpent": 300,
  "avgSpending": 300,
  "bookingsList": [
    "M1"
  ]
},
{
  "_id": 1,
  "totalBookings": 2,
  "totalSpent": 600,
  "avgSpending": 300,
  "bookingsList": [
    "M1",
    "M2"
  ]
},
{
  "_id": 3,
  "totalBookings": 2,
  "totalSpent": 600,
  "avgSpending": 300,
```

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Data Modeling

CONNECTIONS (1) ✕ + ...

Search connecti

localhost:27017

admin

config

day2

local

Welcome: mongosh: localhost:27017

> MONGOSH

> //Customer with multiple bookings

> db.bookings.aggregate([

{

\$group: {

_id: "\$customer_id",

totalBookings: { \$sum: 1 }

}

},

{

\$match: {

totalBookings: { \$gt: 1 }

}

}

])

< {

_id: 1,

totalBookings: 2

}

{

_id: 3,

totalBookings: 2

}

}

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CONNECTIONS (1) ✕ + ...

Search connecti

localh... + > ...

- admin
- config
- day2
- local

Welcome mongosh: localhost:27017 +

> MONGOSH

```
_id: "$customer_id",
totalBookings: { $sum: 1 }
}
},
{
  $match: {
    totalBookings: { $gt: 1 }
  }
}
])
< {
  _id: 1,
  totalBookings: 2
}
{
  _id: 3,
  totalBookings: 2
}
{
  _id: 5,
  totalBookings: 2
}
}
moviedb>
```

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CONNECTIONS (1) X + ...

Search connecti

localh... + > ...

- admin
- config
- day2
- local

Welcome mongosh: localhost:27017 +

>_MONGOSH

```
}
{
  _id: 3,
  totalBookings: 2
}
{
  _id: 5,
  totalBookings: 2
}
> //Genre-wise Booking Count
> db.movies.aggregate([
  {$group: {_id: "$genre", totalBookings: {$sum: 1}}},
  {$sort: {totalBookings: -1}}
])
< {
  _id: 'Sci-Fi',
  totalBookings: 2
}
{
  _id: 'Fantasy',
  totalBookings: 1
}
}
moviedb>
```

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Data Modeling

CONNECTIONS (1) ✕ + ...

Search connecti

localh... + > ...

- admin
- config
- day2
- local

Welcome mongosh: localhost:27017 +

> _MONGOSH

```
> //Movie Popularity with Customer List
> db.bookings.aggregate([
  {$group: {_id: "$movie_id", totalBookings: {$sum: 1}, customerList: {$push: "$customer_id"}}},
  {$sort: {totalBookings: -1}}
])
< {
  _id: 'M2',
  totalBookings: 3,
  customerList: [
    1,
    4,
    5
  ]
}
{
  _id: 'M1',
  totalBookings: 3,
  customerList: [
    1,
    2,
    3
  ]
}
```

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My Queries

Data Modeling

CONNECTIONS (1) ✕ + ...

Search connecti

localh... + > ...

- admin
- config
- day2
- local

Welcome mongosh: localhost:27017 +

> MONGOSH

```
customerList: [
  1,
  4,
  5
]
}
{
  _id: 'M1',
  totalBookings: 3,
  customerList: [
    1,
    2,
    3
  ]
}
{
  _id: 'M3',
  totalBookings: 2,
  customerList: [
    3,
    5
  ]
}
```

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